

ActInf GuestStream 088.1 ~ Abdulrahman: What is the Limit of Our Brains

GuestStream | Abdulrahman: What is the Limit of Our Brains in Predicting Future Events? | 1:49:39

Hello and welcome everyone. This is Active Inference Guest Stream 88.1 on September 17th, 2024. We're here with Hunar Abdulrahman and we'll be discussing what is the limit of our brains. There's a fun backstory here maybe we'll go into. There will be a presentation followed by a discussion so I'm very much looking forward to it and to reading people's comments and questions too. So Hunar, thank you again for joining and to you for any backstory and introduction and for the presentation. So thanks again. Thank you Daniel for inviting me to this nice stream. Actually it's the first time as I told you before we start to do a live stream using this software so I hope the audience will forgive us if there is any upcoming technical problems. Hopefully there will be not. So thanks a lot for inviting me again. My name is Hunar. I have a PhD in neuroscience. I spent many many years trying to learn how our brain works using functional MRI, EEG. So I did my PhD at from, I get my PhD from the Cambridge University and we will be talking a bit about what I did in my PhD and how I try to link my, the concepts that I learn about brain with the outside world with other concepts like the free energy. So I'm actually a newcomer to the free energy principle. I hope that people with technical background forgive us if I have any technical errors because hopefully maybe I'll try to learn and share my insights about what I have discovered from my searching in the past few years. So is it, can you see the slides if I start? Great. So the topic is actually about what's the limit of our brain and I will try to talk about two, two concepts. I, I'll talk about some few concepts and later on we will link them all together. So first, I'll try to talk about the brain as a predictive machine, usually the predictive coding account, then I will try to ease into the free energy principles. And, and then later on, I will try to talk about what makes prediction difficult. I will talk about extrinsic factors. So even if I have, if we have the best learning machine, the best machine learning algorithm, or even if our brain is perfect in learning, there are still a lot of difficulties, basically because of computational irreducibility principle, which I will be talking more about it in the upcoming slides. So I will try to, I would love to start with repetition suppression is a well known concept in neuroscience. So if, if somebody wants to go into the details, what's repetition suppression, I will recommend to you this article by my PhD supervisor Rick Hansen. But it's a bit difficult to wrap your head around the results. So I will try to make it easy. Basically what repetition suppression means, the activity that we record from our brains, we are using EEG, fMRI, or single cell recordings is the for example, if you see a sequence of pictures of elephants, and then later on, you will see something surprising or novel, like the kangaroo image here, the activity of our brain increases increases. However, when we, when you see repetitions of the same image here, like the kangaroo here, the activity decreases. That's what we call a repetition suppression. If you repeat something, the activity of your brain, your brain will not really be responding in the same way it responded the first time. Then you will see something new like this beer here. Because it's novel, the activity of your brain increases again. So, so this repetition

suppression is not only for repetition, it sometimes we call it expectation suppression. So, so even if something is not repeated, but not expected, the activity of your brain increases. But if you expect something, so during, it's usually repetition and expectation are synonymous. If something repeats a lot, it will be expected. If something is novel, it's unexpected. So these results, you will be finding in whether repetition studies or expectation studies. They will tell you, if you see something unexpected or novel, the recordings of your brain increases. If something becomes expected or repeated, the recordings decrease. So that's the repetition suppression. Now, during my PhD, we, we saw a very interesting paper titled Less is More Expectations Sharpens Representations in the Primary Cortex. So, so normally a neuroscientist will be interesting why this happens, why repetition suppression is there. So this paper

proposed a new idea, which actually, a bit surprising for us. So the paper tested, these are grading orientations.

The subject will see different orientations. Sometimes it's expected, sometimes it's not expected. So the results they found is that the activity of the brain decreases when something is expected or, so this was expectation study, not repetition study, but the same. Usually the results will be similar. So the activity decreases, as you see here. I, I, you don't see my mouse. Can you see my, my cursor? Oh, okay.

So the activity decreases. However, for classification, uh, uh, uh, they, they tested classification accuracy. So,

so they have like two orientations. They, so if you repeat something, if you use machine learning to classify between orientations, in the repetition, the classification accuracy is more. So let me see.

Uh, okay. So during, uh, this is the classification, the, the, the bottom panel is about the classification, uh, accuracy. So as you see in the repetition, the classification accuracy increases. And the, uh, their explanation was that because sharpening happens. So I will give you an ex, uh, uh, an illustration.

So this is not the exact experiment, but this, uh, illustration. So for example, here you see the number five. If you repeat that, basically it will become sharper. I mean, the, the neurons, that's the concept that are not, uh, directly representing number five will be suppressed. So when you repeat something or when something become expected, the non-interested activity will become suppressed. So that's their explanation. Basically it's sharpening something, the image will become sharper. Uh, I think this, uh, this is intuitive explanation. Uh, however, we, uh, so during my PhD, we did have a lot of data set about repetition. So two different datasets we have. One dataset was, uh, uh, uh, pictures, uh, of faces, uh, versus scrambled faces, uh, versus scrambled faces were shown to the subjects. So you see here, uh, two faces, uh, similar repeated. And then you see here, two scrambled faces repeated. We also have the data for, uh, gratings, orientations. Uh, it's similar. We do have either repetitions or novel. Uh, so what we found in our, uh, results is same. The repetition, the activity decreases, we saw, but, uh, for the phase data set, we saw that even the classification accuracy also decreases. So it means like when you repeat something, uh, the brain activity decreases, but for the gratings, we all, we saw that the classification increases. So the same result as, uh, the previous group, uh,

which I talked about, they found, they, they, they, they, they explained that as sharpening. So how can we explain this results? Uh, because for the grading data set, we see the classification accuracy increases. So we can say, oh, it's sharpening, but now for the faces, the classification accuracy decreases. So what is that? Uh, so we thought like, it will be very, very difficult without simulations. Just, uh, just, uh, explain the data with our minds. Intuition doesn't work to be quite honest without simulation. So the reason because functional MRI is, uh, even though it's high resolution, it's still very, very low. Each pixel, each, each voxel here represents the activity of millions of neurons averaged together. So, so basically when you are trying to say that, uh, sharpening, it means that then interested neurons will become desactive. We are talking about the average activity of many, many, many, many neurons, and it will be very, very difficult to intuition about that without simulation. So, so this representation will tell you, like, for example, uh, here we, we are, uh, talking about, uh, different, uh, uh, representations of the data. It will be very, very difficult to, uh, to, to conceptualize what happens really in the underlying, uh, the underlying mechanism. Uh, so what we proposed was, uh, we thought like maybe classification accuracy also decreases for local scaling or local, uh, so, so what people think, uh, you see here in the, uh, first panel. So, uh, I just, I'm just wondering where can, can you see the mouse when I am moving it? Yeah, it's small, but we can see it. I'll, I'll also go with my mouse too. Because I have two screens to be quite honest. I, I mean, which screen you see the mouse moving so that I, yeah, we see the mouse, we see the mouse on P1 uniform. Ah, okay. Okay. So, so this screen, okay. I, I just, okay. Okay. So, so five here, uh, if you repeat the five, people usually think that separation will happen uniformly. So you can see the five suppressed with over videos. So sharpening will tell you, no, that's, uh, the suppression will not happen uniformly. It only happens with the non interested, uh, neurons or neurons that doesn't represent five, but we say that even if neurons, for example, the neurons that represent five, even we, if we selectively suppress those neurons, like in the bottom panel, we may still find increase in classification accuracy, depending on how many, uh, uh, voxels or, uh, we, we suppress. So basically this was our conceptual idea before simulations. So how do we simulate data? I mean, how do we go from the neural activity to the functional MRI? Because the simulation is kinda difficult because we are trying to express neural activity in the functional MRI. So one idea is, uh, for example, this is from neural, uh, activity studies from a single neural. Uh, let's say this neuron is very active, uh, for, uh, orientation. If, if the orientation is 90 degree, so you see a lot of spikes here. And, and, uh, the, if you, if you represent that with a tuning curve, you will see that it will have maximum response when the orientation is at 19. Uh, but, but the response of the neuron will be less if you go, because we will say this neuron is very selective to the orientation 90. So the way we simulated that, we, we simulated that using orientation curves. So let's say this orientation curve is, uh, is selective to the horizontal, uh, uh, orientation, uh, but it's less selective to, to the other orientations. What sharpening means, it means it it becomes sharper. When, when we repeat that, the neuron will be more selective to the horizontal selection. So this is less sharpening. This is more sharpening. So that's how we try, we are trying

to simulate that. Now, on the other hand, scaling or dampening, it means like the, the activity will be decreased. The activity of the neuron will be increased for, for the, uh, horizontal stimuli. Now we, we have a lot of neurons, not one neuron. So this was only a representation for one neuron, but we have a lot of neurons. Some neurons are, uh, responding to the orientation 19. Some neurons are responding to, or to the horizontal orientation and some in between, and there is overlapping. I mean, we want to simulate a voxel. We usually some, uh, this line, the activities on this line. Uh, so I, I, I don't know if it is clear or not the, the, the procedure that we are trying to do. So, uh, I hope you interrupt me if something is unclear. So when we sum the activities, we will have a representation for the activity of a voxel. Now, uh, in this case, we talked about sharpening and, uh, scaling, but we have, we may have a global effect. The global effect means everything will be suppressed, everything. So, so this is the global effect. The local effect, it means no selectively a part of, uh, neurons will be suppressed. So this is similar to the sharpening. Uh, we call it remote effect. It means that the neuron that responds to the horizontal, uh, stimuli, it won't be suppressed, but the neurons that respond to other orientations, they will be suppressed according to their distance. So this is a parameter that we can find. And local suppression, it means the opposite, the neuron that is selective to the horizontal stimuli, it will be suppressed. So we see, there are many ways that we can simulate, uh, the, the repetition separation. So, uh, we identified six ways to, to represent repetition suppression. Either we have global, so the red one represents the after repetition, what happens to the neural tuning curves after repetition. So either the global, everything is suppressed or everything is sharpened, become narrower, or we have local, the, the, the stimulus of interest, the effect will be more, uh, concentrated on the neurons that are far away from the stimulus of interest. So for the completion, we actually examined six other models. We call them shifting models, but they are not important because we don't think our brain will behave like that, but, but just for completeness. So for those who want to do a mathematical modeling, uh, it's just like for, here G is the tuning curve, which is represented by X and μ , which is the mean of the tuning curve, and σ , which is how, how much narrow is the tuning curve. And then we multiply it by a factor, which is C , that's the suppressing factor. So, so in local scaling, we suppress the tuning curve. In, in, in sharpening models, we suppress the, uh, the σ , or the, how much narrow is, and then in shifting models, we, we shift the, uh, tuning curve. So for more technical details, you can actually go to the, to our paper and see more details about how did we simulate that. Uh, uh, and as I said, we have two data sets, the phase data set and the grading data set. So what we want, we want to see if our sim, if we can use our simulations to, to match our findings, uh, and actually, uh, using functional MRI, we actually identified some other features. We didn't only use, uh, repetition, suppression, and, uh, classification. We identified, uh, six more, uh, data features, and you want to see if we can actually use the simulations to, uh, explain that. And the results which we found was very, very interesting. So these green, uh, circles, it means that that specific model. So remember, we, we, uh, we examined 12 models in total, 12 different models. So if it is green, it means this model can explain this functional MRI

feature. If it's red, it means it can't explain. So for global scaling, you can see, it can explain a lot of features, but except, uh, BC, which, uh, I didn't have time to, uh, these are functional MRI features. You can see, this is the data and we are trying to fit it. Uh, so the interesting thing is, you only see local model, local scaling, can actually explain all the features. So you don't see any red circles here for both datasets. So it can qualitatively explain all the features that we saw in our data. All the other models, they, they can individually, they can explain individual features. For example, here, sharpening model can, uh, explain, uh, repetition separation, can explain within correlations, can explain between correlation, but it can't explain, for example, the features of sharpening in our face dataset. So the short story long, uh, as the, as the long story short story, we find that local scaling best represents our, uh, the, the findings of our data. So the, the, basically the local scaling, it means that the best model is the model that actually predict when we repeat the neuron that, uh, that are selective to that feature, it will be suppressed the most, not global and not sharpening. The, the suppression of the neuron that, so basically what that means, if you see, uh, a horizontal orientation, the neurons that are specific to this, uh, horizontal orientation, they will be suppressed more. The more selectivity to that, they will be more suppressed. The, there are two reasons we like the results like that. So basically later on the paper, which we actually, uh, introduced later, they did similar, uh, uh, simulation to us and they actually found similar conclusions that, uh, these model, this neuron is, uh, this model is best explains our dataset. So the, the reason we like, uh, local scaling or dampening, uh, it's, it's become, it's because very parsimonious, it can be explained easily without like instigating a complicated explanation. So basically it just means that a synaptic fatigue. So when you, when you stimulate a neuron, when a neuron is stimulated, it will get tired and then it will be more suppressed when you repeat something. So this is a parsimonious explanation. And the second reason why we like this result, because now I'm trying to connect it with predictive coding literature and with the free energy principle, but first with the predictive coding literature, we think this result is more in line with what, because you see in, in predictive coding literature, you usually see this image a lot. You will say, oh, you have a stimuli and then you have a prediction, the lower upper ribbon of your brain, try to predict and there will be an error and you have a prediction error. So the question is not what this prediction error represents. We think that these results will tell us that the prediction error is the unexplained data. So it's, uh, uh, nowhere says that the prediction error is the noise. It's usually the non-explained stimuli. So the part that you can't explain. And that we think is actually more in line with, with the suppression, with the, with this model. I will actually come back to this, uh, later. Uh, but, but, uh, first I want to talk about, so we established that our brain is a prediction was machine and, and the results in our functional MRI actually supports that or brain seemingly it's not interested in expected stimuli because it has predicted that it's, uh, our brain trying to predict the non-expected. That's why more activity is being exploited for novel and non-explained, uh, stimuli. So what's the enemy of prediction? So if our brain is a prediction machine, what makes, uh, what's, what are the obstacles in front of, uh, our ability or our brain to predict things? So, uh, I will tell you a story that I faced. I, I'm usually interested to train MRI, uh, at some

point of my, of my life, I was trying to train a neural network to predict a Mandelbrot set. And the reason why I thought Mandelbrot set is more interesting than language predictions, predicting the next token, because from Mandelbrot, uh, Mandelbrot, you actually know the ground truth. The ground truth is actually, if you copy this JavaScript line into a, uh, JavaScript, it will produce a Mandelbrot. It's very, very easy. It's a few lines of code, uh, just one equation. It will produce. And then when you change these numbers, you can zoom in to this Mandelbrot set. So for Mandelbrot set, I know actually what's the ground truth. It's not like the language. I, for language, I don't know what's the best ground truth. Is, is my sentence is the best? Is my, is your sentence? Which phrasing is the ground truth? We, so, so the, it's very difficult to identify the ground truth in language models, but for Mandelbrot, we know what's the ground truth. And that's why I was interested to train neural networks to predict Mandelbrot set. Uh, so the surprising thing for me, I thought this will be easy because Mandelbrot has a lot of regularities, a lot of, uh, places where it repeats. So I thought this will be an easy task, but I was shocked. I mean, even when I used a neural network, uh, with very, very big size to predict different parts of one, so you can train a neural network to do, to predict a specific part. That's easy. But when you want to generalize, it will be very difficult. The reason is because I was underestimating Mandelbrot. There are, it's true that there are a lot of regularities, but the more you zoom in, there are, there will be always new features. It's mind blowing how this simple line of recursive, they will always present you with something new. That your model or the neural network, no matter how good, how good is trained, it can't predict any, anything. So, so, so that, the, the puzzle for me, actually, uh, we know that recurrent neural network are too incomplete. So, it is possible that you can train a neural network to simulate that code. But in that case, the neural network will not help you in prediction. It will just simulate the same code for you. The, the, the main hope that we have with neural network is that we predict ahead of time. We predict two steps ahead of time or three steps ahead of time, not, not just recursively, uh, predict one step because that's what, uh, which I will come later on, on computational, it's the first of that.

Uh, so, so before coming to computational irreducible, so this is my first time when I saw there is a challenge in prediction. It's, it's, it's, it's very difficult, uh, to predict these intricate features, never ending emergent patterns. So, one, one of the reasons, uh, this is phrased by Steven Wolfram is the principle of computational equivalence. So basically, if you manage to train a recurrent neural network to simulate the same rule that, uh, that predict, uh, that, that actually generate mental broad, it means that in the underlying leap at the underlying level, the best time, they are doing the same thing. They are doing the same computation. So you, uh, basically according to this principle, you cannot simulate a rule with another rule that are more efficient. That basically, if you, if you simulate them at their most basic level, so you can simulate rule a with rule b, and then rule b can actually predict two steps ahead of rule b. That's not possible. So this may be difficult to understand, but the way Steven Wolfram did it, he explained that in his TED talk. So, uh, he actually, so basically the, in, in his TED talk, he was like, uh, I'm paraphrasing. He was like saying, okay, you have this simple rule. So this is just like a mental product. You, you just gave this

rule to computer, and then it will produce for you a pattern. These are step-by-step patterns. And then it will tell you if, if you could, is it, is it possible to predict ahead of time just from knowing the rule? I mean, you know, the rule, uh, just, just like, we know the theory of everything. Can we, uh, can we actually predict ahead of time without stimulating the intermediate steps? And I actually don't know if it's true or not, but I believe that because Steven Wolfram have spent like his decades of his life trying to do this. And he says that this is not possible. You have to, you have to compute all the steps. You have to compute all the steps. And he even put a prize for that. The reason why I believe computational irreducibility principles should be, it must be true, because it's, uh, in essence, it's similar to two other, uh, principles, Godel's incompleteness theorem and the whole thing for the problem. In, in spirit, if you think more about it, they, you, you will see them that the, these are all talking about probably the same phenomena, but in different, uh, uh, so what I learned is that complexity is easy to generate, actually. Uh, it was a very shocking thing for me, but I have been simulating these simple rules and then they, uh, for a while. And I, I, I, they always surprised me what they are can produce very simple rules. You fine tune them and you can generate, but the problem is it's hard to understand. And this is because of, um, that principle which was, which was formulated by Steven Wolfram, which is the says computational irreducibility principles. So basically let's say we have two systems system one and system two, and then system one tries to predict system two. So if, if system one can predict system two with less computational power, then we can say that system two is reducible and this system one is analogous to our brain and system two is analogous to the environment in the world, which we are trying to learn about. So what Steven Wolfram that he examined so many different rules, uh, and, and the reason why he ex, uh, examined actually cellular automata because of their simplicity. So what he found that is each rule are, uh, uh, producing unique patterns.

So, uh, actually by, by looking at these, uh, patterns, you can see what computational equivalence means. It just means that you can't, you can't stimulate, for example, this, uh, pattern using another rule. Now there are some rules that are faster than itself. That's what I mean. Now there are rules that are Turing complete, like the rule, uh, 110, but even if you manage to initialize the rule number 10 to produce another rule, it will not produce it faster. I mean, yeah, you can't use it two steps ahead of time. So, so some of these rules will be, uh, trapped into repetition and some of these rules will never be able to predict what comes next. And I thought there is another phenomena like that. It's, uh, the irrational numbers versus, uh, rational numbers. They are actually similar to that because we have irrational number and then their definition is if you go, what's the definition? They, they will never trap into repetition. So we have pi, which is a very simple rule. It's just less certain parts divided by radius. It's a very, very simple rule. If you simulate that rule, it will always give you a new pattern, a new number, uh, the same for the square root of two, three. So this is amazing. I mean, I, I don't know why I didn't thought about that, uh, before Stephen Wolfram talking about the last, uh, uh, silver automatic thing, because I thought like these are actually have the same

amazement. So if you have a generative, it just, and, and actually we usually talk about, uh, universality, Turing complete universality. So they will tell you, uh, if you have an infinite theme and if you have a reader, a writer, and specific rules, you can compute everything and they will call that, uh, Turing complete universality. Now, according to Stephen Wolfram, he actually revised the, what, what does it mean by computational universality? So, so he usually, uh, explains that in the term of cellular automata, but I will try to explain that in, in the term of rational and irrational numbers. So it's actually a bit mind blowing, but we, we, we usually don't think about it like that. So let's say pi, uh, that's the definition. A definition means this rule will never trap in trip, unlike the rational rules. So the rational, uh, numbers, if you divide them at some point, they will repeat, they will repeat, they will never give new patterns, uh, but, but irrational numbers. So that means if you, if you compute pi for infinity, it is possible that at some point you will find the same pattern that the square represents the square root of two. So, so if you, if you compute pi for infinity, it's possible that you can find a space of one, one, one, one, one, but, but it will never trap into that option as, at some point it will, it will change. So, so that means in every irrational number you have, uh, the other irrational numbers embedded in it. There is no way to say this is not true because if you say this is not true, it means that these patterns will be trapped at some point into a cyclic repetition, but if they don't trap, they will always give you novelty. And that means, so someone can tell you pi is too incomplete. Why? Because he says like, if you, if you, if you keep computing pi for infinity, at some point you will find that it computes my genetics, the same sequence that if you represent it as a number. I, I mean, you can say this is not true. Can you, I mean, do you have a comment, Daniel?

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is possible is because two things memorization and abstraction the reason why i didn't put memorization before abstraction because memorization is not very without abstraction memorization is not useful because memorization it means if something repeat then if you memorize it later on you just catch that and if something has that it will be computationally very easy to predict but that will be possible if we if we live in a computationally really reduce of the universe if something repeats memorization will be great but if we live in a computationally irreducible where there is always emergent pattern memorization is doesn't help the reason because you should have infinite size memory to predict to to store all the all the emergent patterns you have to have abstraction with memorization it means just like uh so remember that uh previously before ai people store pixels inside the computer just using memorization but what neural networks do is does is they memorize but not just uh using that it will abstract the information before memorization so so so you just don't memorize the data like itself you you have to find a way to find the useful uh so so that's the formal definition i actually gave that to jack and that's just the which i like it actually more than the wikipedia definition uh is the process of reducing complexity by focusing on main ideas and acknowledge specific details so that's the only way that we can beat the computational irreducible it's just we pay attention to what matters for our survival so you can see i'm trying to actually reconnect back to the active inference so there are many ways to abstract there are many different algorithms but i think the easiest one is uh that's what they usually introduce you in category theory they just say oh we we just uh gave us same name to everything because uh so let's say we have bob ali joe and if you even want to do statistics or if you want to find any any relationship we just need to

group things according to their similarity so similarity is actually a very complex topic it's not just easy to say according to similarity but but they will be right now so first thing is actually simpler because course training you group things according to proximity proximity is easier to define than similarity you just say if things they have a uh equidistant distance then uh specific threshold you just group them so course training actually used everywhere i i'm surprised the course training is not not so much well known but it's it's it's used everywhere another thing which may be more familiar to your group is markov blankets i think markov blanket is also a more sophisticated way of abstraction uh the reason why micro for for example here you see these notes the parent notes the children notes and the partner notes they are and then internal notes so in the markov blanket it becomes stable you can just treat this as one entity so just just like another way of abstraction so we say anything inside this membrane is just one entity anything inside this house the house is one entity with everything that happens inside i don't care about only when somebody comes outside the house i should care about but when they are inside about i should just look at the house so my my point is that the stable membranes can abstract things for us uh linear regression is another way to abstract information but linear regression is just a more sophisticated way of course grading because we are using a weighted in course training we just average uh the nearby information but here uh we are we are doing uh uh weighted uh regression or weighted the summation up uh so we have neural network as well neurons the most sophisticated way of abstraction so far uh so so coarse graining let me just give you a bit about coarse graining the importance of course training so i mean it's self-explanatory you just average the nearby pixels or the nearby uh so so what i think about uh talk about uh why course training is important there is a simulation called particle life simulation i have a video about that a detail but let me just i don't know why this particle simulation is not more uh familiar among the scientists uh it's it's actually a more interesting extension of the game of life i don't know who invented that but there are old videos uh seven years ago doing these particle simulations i didn't have the quote back then when i wanted to re-stimulate that so i improvised the quote and i was shocked the reason why i i just wanted to see this is to see this is to see this is to see this is to see this is to see this is to see this

lines of code. These few lines of code just represent particles attracting to each other according to Newton's law, something similar to Newton's law. So the inner loop here just says if the distance is this, you have an attraction force or a repulsion force parametrized by g . So the g parameter actually determines the attraction and repulsion forces. So just less than a page line of code. And when you have this, you can simulate this one. For example, you can specify how green particles attracted to each other and how red particles attracted to it. So just few lines of code and you will have so many interesting patterns, so many interesting patterns. So we have made a C++ software about that. It's available on our GitHub. So it's just amazing how complexity is easy to us to generate. So you just have these parameters are just attraction and repulsion between different color particles. So you just define how much red particles attracted

to red particles. And then you let the simulation run. And you will, as we find 20s, so maybe, maybe if you are interested, I can give you a live demo. Is it possible? Is it okay? Oh, yeah. Okay. Okay. So if you go to the GitHub, there is actually a link, a JavaScript. Let me reshare the screen. Okay. So you can see the screen. Okay. So if you go to the GitHub, winner4321particle.live. And then we have a downloadable version, so you can download it. And we have an online version, a 2D online version. So let me just open that 2D online version. I hope you can still see it. Do you see it or should I?

Yep. And I'm also pasting it in the chat so people can go to it.

Okay. So the interesting thing is, so you see, I will just, so here are the parameters. So I'm trying to change the parameters. But even if you push the random button, so here is a random button. So this random button will just randomly revisit the attraction and repulsion force. So even if you push the random button, you can still see interesting patterns emerging on the screen. So let me zoom. Yeah. So, so I'm, I'm now just, so you can see, I'm just now pushing the random button and you see cellular things like very, very interesting things coming to.

So, so we will come soon to Markov blankets. You'll see that Markov blanket is very natural. It's something when you have local rules, that's what Carl Probst usually says. When you have local rules, there will be always a Markov blanket. So because we have local rules, but if it global, so the attraction and repulsion rules, they have a threshold. So you usually say this particle can attract this particle, but there is a circle. It can't, it can't attract very far particles, the same power as nearby particles, just like in real life, just like the gravity in real life. So you can let these systems evolve. They will produce, but what interests me is even if you push the random buttons, the random, uh, uh, this random push it and, and you will have interesting patterns.

So now you see a bit, a bit stability. Now you can see a stability, but this stability will not go forever because of computational readers. Sometimes something comes to disturb the system. So now they have a Markov blanket, they live happy. They, they, they have their internal system, but at some point you see this, ah, okay. Now the system is disturbed. So they, they, they live for a period of time. They have nice Markov blanket, but there's something outside happening.

That's the, uh, the halting problem is, could you look at this at a given time, all the rules, you know, every single particle state, all the rules, could you say what the final end point will be or what will happen in a certain number of steps. And you might have a heuristic, but you can't know without actually going through the procedure.

!

Without actually going through it would mean simulating the system. I mean, if why, if I want to predict the future, I have, I should have a supercomputer that is faster than this. I should put the initial settings and the supercomputer can simulate the system faster than me. But, but if I am living inside this system, I don't have this leisure to, to simulate the system faster than a cell. So, uh, one other, one other interesting piece there, which you've brought up a few times with like having a more powerful computer somehow outside the system. So as you start to look deeper

and deeper into futures, it's almost like the procedure has runtime. And like, it's just whatever complexity class it is, it's just one or X. But if you start looking further and further ahead, there are more and more adjacent possible so that you actually, it's like trying to run up a wall that's getting exponentially steeper because the, the faster supercomputer, even if it's, it's a million times faster, but then if it starts looking several time steps ahead, each of those time steps have a bigger and bigger adjacent possible. So eventually that 1 million fold increase gets eaten up pretty fast. Yes. Yes. Yes. Yes. So, so that's why I think this part, I actually, Carl Frunston is this paper that it has a primary also, but they usually, I think it uses MATLAB and then, so I mean, people nowadays usually use more sophisticated tools that have nicer visualizations and, and you actually couldn't find these interesting patterns if you didn't have a real system. So, so when I was coding this system, it's impossible to find interesting patterns without having a GUI, a GUI system to explore easily in real time. Uh, and that's what made us, uh, discover so many, many interesting patterns. So now there are other systems that are similar to it. Actually, I think there is a system called linear, which are smoother. I think it's also very interesting, but the reason why I like that it's very naturally connected to Markov blankets and, and active energy. So, so I will actually now go to, to what we can actually elicit from these systems and how can, how can we, uh, do abstractions and how can we reconnect what we have learned to our findings that I explained in the beginning of the video. So let me share, reshare my, uh, uh, my screen again. Uh, so I'm, I'm sharing the slides again. Okay. So, so you can see the slides again. Yep. Okay. So you saw like, like these parameters, I found these interesting patterns, but somebody can reproduce it. If I download the software, the C plus plus one, not the online one, because the C plus one, one is more powerful. You can simulate more particles without, uh, I mean, uh, the computer can handle it. So, so you can find amazing, amazing patterns, just, uh, and this is just four particles, four colored particles. I mean, you can go five colored particles, six colored particles. Okay. So why, uh, so my point is how, how, how, how, uh, coarse graining can help us, uh, uh, in, in this case. So let's say, uh, I mean, this bug shape, to be quite honest, I actually produced this bug shape from, from, uh, so, so, so this is also not the profile. The profile is just for the illustration, but this bug, which you can see, this is from the particle life simulations. This is, believe it or not, just, just from these rules. And I accidentally have, uh, to me. So green attracts to green with this power, green attracts red or repulse red. Uh, I, I forget whether the negatives of attraction or the positive, but, but these simple rules of attractions, it can produce you this flying shape, a bug-like flying shape. So this is, uh, some screenshots, uh, how, how it, how it flies around. So my point is that, let's say this frog is inside this system and it's a predator. It wants to predict because without prediction, you can't catch this bug. You can't catch this bug. But the frog is inside the system according to computational pre-reduceraally, it can't stimulate the system faster than itself. So it can't know. The only way it can do is does some, some way of abstraction. And the easiest abstraction, as we say, is coarse graining. So if, if we do coarse graining, so here I did coarse graining.

This is not a simulation. It's just illustration to be quite honest, but I think it's trivial to understand. So, so now the number of particles becomes 70 particles. So if the, if the brain of the frog, it just samples the, the, it, it just didn't sample all the 600 particles. It just samples based on locality. I mean, local things, sample them and then, uh, in the future, I mean, uh, repeatedly, local things will be aggregated together. You will have a coarse grained image. And this coarse grained image is very, very easier to simulate for the frog. Now it deals with 70 particles instead of, uh, 600 particles. And that's why maybe even if the frog is inside the system, it can actually predict the simulation. Of course, it won't be exactly the same simulation because we are simulating a coarse grained world, but it, it should, it, it will be close enough. It will be close enough for the frog to survive. So, so if you do the real simulation, the, the bug may go this way and then go this way. But if you do the coarse grained simulation inside the brain of the frog, the, it may predict a different path. But if the path is close enough for the frog to survive, then that's it. So this is what we come here into a very interesting conclusion. Living things like a frog will value accuracy and will value efficiency. It values both accuracy and efficiency. So you see, we, we, we are closely intuitively coming to the free energy principle, which tells you, which tells you, it's, you have to minimize surprise and complexity. That's what the, the equation of free energy. So, so it's the same. You, you, you can say, so let, let me go. Actually, I, I explained that in the video that coarse graining is a very underrated concept. It's like, it's used in physics, in chemistry. Do you know there is in physics, there are so many theories under the name of effective theories. They use coarse graining to explain so many things. And usually you do not see physicists telling people, oh, we predicted that, but we use coarse graining. I, I saw that this is like hiding a different fact, because we usually talk about uncertainty principle and things like that. Okay. But you use, you use coarse graining. Coarse graining is part of uncertainty. So, so in this simulation now, it's true that we can predict the bug, but now we are less certain about the individual particles because we did coarse graining. But, but that was okay for us, because the aim of the frog is to predict the whole system, not the individual particles. So I'm actually surprised why this is usually don't explicitly say, hey, we use coarse graining. So they, they used everyone to be quite honest. So many theories of physics depends on, they, they just use all the, even if only when you go to the details, you will find them. They have coarse grained the system before doing the simulations, whether it's a fluid system, a particle system, anything. Well, well, I think everything is coarse grained. I don't think we can measure anything precisely. So coarse graining is there, whether we like it or not. Okay. Markov blankets. I think Markov blankets is well explained. Carl Frost have many, many videos about Markov blankets. So Markov blankets is a different way of abstracting information. So the nodes that are surrounding, let's say we want to predict x . So if we want to predict x exactly, we have to know the, we have to know the position and the velocity of all the particles in the system. But according to Markov blanket, you don't have to. If you know the state of the Markov blanket, you only need to simulate the state. But the problem is how do we know the state of the Markov blanket? We know, we need to know the state of their Markov blanket. But how do we know the

state of,

so that will defeat the purpose? It means that we have to simulate the entire system. But as I said, luckily, these Markov blankets are sparsely connected. I mean, you sometimes see the world, the reason to be quite honest, I'm still not sure why, why are world are sparsely connected? The only reason that comes to my mind is because we have different force ranges. We have nuclear force, which is very very short range we have uh electromagnetic force which is wider range we have gravity force and then because there is a speed limit of light if you combine the speed of limit of light and the the range of forces you can have various ranges of forces just like just like in our simulation so just like in our simulation so we can also say that the particle simulation is also sparsely connected because we have simulated different ranges of forces that's why close by particles so if they will arrange sooner to to produce lumps so that's why i think markov blankets are possible stable markov blankets are for because of of sparsely connected system why sparsely connected system because of different range forces that's my my explanation for that and this makes prediction easier now the reason why i think this makes prediction easier uh so i will give you this example uh last example of particle simulation because to be quite honest the reason why i like particle simulation is it gives you intuition about a lot of things so one of the intuitions i usually pull up uh steven wolfram's video we usually say oh i understand entropy now i think entropy is the feature of observers because we observers are computationally limited that's why entropy is there i was usually think what the what the hell he is talking about it's a bit so i finally have aha moment when i was trying to understand how entropy works when watching these particle systems so so let's say this was the initial uh state of the before we start the simulation this is all the particles if if somebody sees this system it says this system have half high entropy it's intuitive i mean even if we don't specifically define what's entropy here it's this system is all sparse around different colors everywhere so if i tell you can you predict what will happen to this particle here at the middle which encircled by the where does this particle go uh i mean you don't know i am sure you don't know the reason you don't know even you have the root it's just your brain can it can it simulate all the particles inside your brain at the forces you can't you can't figure out how what's the direction of this particle will it go this way this way but if your brain is unlimited power i mean if you can compute all these particles right away there is no uncertainty you can tell me i know this particle will go this direction but the reason you can't tell me because you can't compute all these particles now as the system evolves like you you let the system evolves now we have different entities with different mark of blankets they are stable so you have chunkers so now your brain can meaningfully coarse grain so so you can easily course when you can say oh we have four systems and these four systems they attract everything so now if i tell you can you tell me where this dot goes according to the attraction rules i think now you can say you can say oh this this dot is very close i think it will go down so the entropy in your brain is increased so the entropy i mean the way that the uncertainty is decreased sorry because now you can predict where this dot will go and the reason you go because now your brain can compute what happens because because your brain chunked coarse grained

all these particles together as one entity that's because of the markov blanket or because of these stable blankets which which encapsulate these entities you you treat them as one entity and the system is now become more predictable

so now you can tell these particles will go downward and uh and you can say this will go from high energy to low energy so so

basically you can use these particle systems i actually didn't do that it's in my plan to do actual simulations not just intuition about these systems so this is gibbs free energy in which uh carl christen actually derives from the concept of pre-energy in the living beings it just relates entropy with uh enthalpy so so we know that entropy of systems increase but local entropy because of these local forces local entropy decreases but the entropy overall entropy decreases because particles will go down and then uh the energy uh as a light will go into the universe basically these simulations are so simple to capture all the details but all the details the global entropy either stays same or increase but the local entropy decrease and that's what self-organizing systems like us living beings will try to do they will go to that so so what carl christen does is generalize this concept to living beings and uh to be quite honest i don't know the details that he do that but he usually use these sequence langevin equations with full body attractors fokker plank equation mark then he comes to the market blanket once it reaches the market blanket it can connect to the living world bayesian world and then and then we have so these particle world can be connected with living world that's what the carl christen usually does and the interesting thing it's the same objective so that the high level equation in which that carl christen free energy people uses it just you have uh sleeping systems want to reduce surprise and not reduce complexity and if you remember we told you because of computational we want to be efficient and we want to increase the accuracy so we reach the same intuition without driving it just because we have computation we have to be efficient there is no way around that so this part and that's what makes because a long time ago i didn't understand what's exactly the difference between free energy principle and the predictive coding so the main difference is that the pre-energy principle also adds efficiency it says that we try to predict but we also try to be as efficient as possible as a simple outcomes principle way or the least action uh way so it may it tries to reduce the complexity and i think the the the reason why we want to reduce the complexity is because we have a computational resolution principle so now we come to active in france because according to active in france there is a trick we can do we can not only predict the system by learning about it we can actually be participate in the system so so you either learn from the world uh which i call it the bottom-up system or teach the world teach the world that means you you are trying to change the world so just like if you go back to this to this particle it's just like you see so this particle either it attracts the entire system to itself or even the particle goes to the bigger system so we individuals also do that either we are trying to change the community or either we adapt to the community and usually have to do the later the majority of us will be just slaves like we will just say we can change the world usually but i think the analogy is very easy to to see uh so so there are a lot of questions uh so i i usually usually this question comes up if we try to predict the systems why we just don't go to the dark room uh i see i i also see the answer to this

from the computational irreducibility principle because computation doesn't let you there will be always emergent factors so locally you can do that as an individual but collectively we can we can do that because the system will always change uh this is analogous another example which i like i don't see it's used why we don't use fascism because fascism actually makes the system predictable everything behaves according to some order uh i mean we can use a less uh inflammatory word than fascism actually people do that people try i mean you can see religion as a way to make everything predictable so if you are if you follow a community a strict community they will tell you you should behave that you should do that the whole day will come down so everything will become predictable and that's why these systems sometimes are appealing for people because it encapsulates the idea but the problem with that again computational doesn't let you to live that forever there will be you will be trapped on a local maxima and that and you you can do that you can live in this predictable system for years for thousands of years but the outside world will always change will always change and at some point these changes will either affect you collectively or individually others that's why we have to change so so well that's why we have curiosity we want to learn i mean we can never uh so so according to the simulations that have that been done in free energy principle even these particle systems they are actually modeling the outside the reason why they did that so if you if you try to uh follow the activity of these particles they you will see they are correlating with the outside if a change happens in outside you will see a change happens in the inside so as if the inside of these entities they are trying to model and the reason why they are trying to do if you if you see that the mark of blanket just actually just looks like a neural network so it has sensory nodes it have inter internal nodes it has active nodes so it's just like a neural network it's just like how how it's trained so either use evolutionary technique to train it the the some blankets survive some will not survive so we are using evolutionary technique to train it or you can use back propagation or you can use the technique that happens inside our brains ebn learning although there are a lot of unknowns how how the learning happens in our brain but but the thing is same so so nested levels of markov blanket you can see it as a nested level of neural networks it's same it's same level of abstraction uh okay so so that's what makes us that's what i'm saying we can beat the computational irreducibility because so all the brains are local they they can't compute the entire system but collectively the the lower brains they abstract the information they don't send the entire information to the upper the higher level brain they abstract it away and they so that the higher level brain even though it's computationally very limited it has the same it can actually simulate the system because it's abstracted it's coarse grain uh so i i it's usually a balance between top down and bottom up so basically what happens is that the inner levels will be more predictable but the outer edge will always be less predictable or the higher levels will always be and we are i'm usually saying cautiousness leaves at the higher people at the edge of uncertainty so we have subconscious everything is predictable like for example if a fire here my hand will go away the gland ganglions in our spinal cord will decide even before goes to our brain the decision so predictable things are happens in your subconscious the unpredictable things happens in our conscious uh okay i think uh i think i have tried to explain all the important things that i want

to do but but just before finishing the uh i want to just reconnect this concept uh back so so far we are talking about how do we do abstractions we generally the best abstraction as i said we have neural networks and help neural network using back propagation and how back operation work we really don't understand even people who who who uh produce chat gpt they say we can put the the system for you but we don't know

how back propagation will decide how abstractions are formed so i i mean few years ago i came across an algorithm which is which actually uh drives multi-linear regression i i thought this is beautiful because it it matches with our predictions without with our simulations very very well the simulations that i did in the beginning in the beginning so i don't know have you heard about lattice filters lattice filters

yeah go go for it though continue yeah so lattice filters are actually

they used all the way in kalman filters and is signal processing filters and i just don't know why they are less famous in the machine learning community they are more if you go to the signal processing community you will see a lot of talk about lattice filters and kalman filter and things like that for me lattice filters are actually very very similar to how uh the brain is depicted how because the predictive coding system will tell you we have predictions and we have prediction errors and they match very very neatly together and these lattice filters are very very easily derived so if you have two groups for example let's say we want to see which group are more liable to have a blood high blood sugar we have obese group and

less obesity and usually in scientific tests we have more variations and these variations will affect our results so the best experimental design is to remove these variations uh that that's what we say control design but sometimes you can't do that because there will be always variations so many multifactorials that even not aware of it so what you can do that you can do that post hoc uh so so let's say we here x we we want to predict x uh these x's using x_3 and what what's not predicted is we call them prediction error so the prediction error will go to uh to the upper levels and then we use the residual errors to predict themselves the other remaining errors so you do that until you end up with the last uh prediction error which represents this picture in in this case obesity this is where you can actually find the relation with obesity and uh why the factor that you want to test and if you do that sequentially for a sequential data you actually end up the driving the lattice filter again which which i talked about here so you can you you will drive lattice filter and this lattice filter is actually very very similar to how a prediction error works and this this way it's similar to what our findings in in our paper when we say the expected stimuli will be suppressed so these lattice filter will also tell you the same thing it will tell you uh in multivariate linear regression you are trying to predict the stimuli the prediction error which is not predicted it will go to the upper level and then the the outer context will try to predict that and uh you will go to increase the context to predict the remaining errors so i thought this is a really an interesting connection but these lattice filters are not used in machine learning the reason is because mostly they are linear systems there are ways that you can make it non-linear but they are because they learn very fast they are very very sensitive to noise so if you have a small noise they will just learn the noise too and that's why that's why i think people in machine learning system they actually go to

use optimization methods using back propagation because in back propagation you have a learning rate which you can reduce it that means even if a noise comes it will not affect the entire system so that's why my only explanation why people don't use this method but i like this method because it connects our brain the predicted coding error and it is very similar to the uh findings that we have in our paper and and i think it connects nicely somehow to the free energy principle so uh thank you a lot for for uh listening and i'm here if you have any questions uh uh repetition suppression uh repetition suppression in your phd then what what what journey has brought you to making the self-organizing particles and all the cool youtube videos oh thank you uh well repetition suppression uh it's uh my phd supervisor is actually uh a very uh big name in repetition suppression so he he have a lot of reviewed uh review papers on repetition suppression uh and uh uh and i found this is interesting uh my beginning phd i was interested in episodic memory because that was my master's degree uh uh was about episodic memory uh but but later on i thought there is a very nice connection between repetition suppression and uh predictive coding literature uh back then i wasn't aware too much with free energy principle but more aware with prediction uh predicted coding literature

so i thought because there is this connection and and and it was always surprising to me uh the phenomena of repetition suppression it just means that when something becomes very expected when when you when you finally predict something your brain will go idle so usually i thought in the beginning it's counter-intuitive because usually you think oh if something becomes very predictable you should your brain should be more active uh but but that's uh interestingly not true your brain is uh is usually but what i think is is more active at the age of uncertainty when when you because because let's be honest if you if you see noise it's completely unpredictable but at some point your brain will go idle and interested so we uh our paper was not about that but i am sure if there are papers uh it's a u-shaped curve so if something is very predictable your brain will be less interested less active and if something is completely unpredictable again we we are not interested in it so i think that for some reason i think it may be connected to consciousness i am not sure about my my studies was all about consciousness but i think repetition suppression is probably the core the closest phenomena that we can objectively examine and in some

ways connects because because subjectively uh when something becomes predictable you also feel it's boring and you feel sleepy so so so the objective findings that we see the activity decreases also correlates with the subjective findings that we actually feel and because these are two connected i think it's probably the closest uh objective phenomena that we can test and somehow we can correlate to consciousness

yeah i mean the example of the just a random pattern that could be intelligently coarse grained and you go okay

frame after frame the coarse graining of this whole picture is just gray so then the coarse graining is unsurprising so then even if the pixel level details are still surprising you've reduced or bounded surprise at the slower or the deeper layer so again that becomes not interesting oh wow that's actually a very

so so so i actually didn't thought about it like that but the coarse graining idea if you combine it it will it will just reconnect back to the morex that that's very nice actually i i really liked how you connected the coarse graining with the the markov blanket a lot of work has connected neural networks and the loss function that's used in training the neural network with the variational free energy and then in the neuroimaging setting coarse graining like you brought up it's a really simple heuristic because it's just spatial proximity you don't need to cluster things by their functional similarity you just use the anatomy and then it's interesting to think about the markov blankets as coarse graining in a causal space like if you had um 10 10 different uh sub procedures but then you say well one procedure one two and three are like one unit and then that becomes like its own thing but procedure one might be anatomically close to seven but then one two and three are causal and so then it's like building these parallel maps from your phd it was more grounded in the spatial and the temporal and then the markov blanket abstracts that into a causal but then abstracting away from space and time is where we get patterns and that is how abstraction helps us predict the future exactly oh god this is interesting i think i think i think of course training is just like a door for understanding so many things and i usually i i sometimes interested in physics because even in my high school i was interested in physics i don't have a formal education in physics but i am very very interested and i am very very interested in the phenomena of uncertainty principle and and bills phenomena and i usually i'm afraid to ask this because they usually get angry is there a relation between course training and uncertainty they usually tell you no and uncertainty in physics is fundamental they i i don't know whether they may be there are maybe physicists outside of life physics like for steven wolfram or uh some other physicists they they have more deterministic uh uh view of what uncertainty means or what randomness means at the lower level but to my experience usually when i go discussing reddit they will get angry they will say no you don't understand anything about the underlying atoms they will just go random it's just a a true random generator well i i don't know who will get mad about what but in a way the movement from quantum scale physics to classical physics is like coarse graining it's like saying well if you go to one atom it might be difficult to know what vector it's going to move in next even if you knew the temperature so maybe that's near the limits of of um a true ontological uncertainty but then if you have a ball moving through space you can get a really good prediction on a slower larger thing and then that's classical mechanics where you don't have the same uncertainty you can just assume this basically infinite precision but then quantum is where you can't assume infinite precision and people debate whether that's ontologically how it is at a certain scale or whether that's an artifact of our interfacing with small things but regardless things that are about as big as we are it's kind of like these are like pebbles that are about as big as we can pick up those are the things that we can think about that's what makes sense to us and then those are the classics and then things that are um more challenging to abstract or predict they're nonsensical or they're they're outside of our our scope of understanding yes um but but from earlier how did you get from the repetition suppression to to working on what you're doing now and and making cool videos uh actually uh particle uh making videos uh i i i as i told you before i actually

learn more when i make a video about the topic so i call that active learning uh there are so many topics that in my life i want to try learning it's just like because i see many intelligent people they say oh this this topic is very important this topic so right now in in current situation i'm trying to learn mona and i i i look at many videos i don't know what the hell they are talking about but because so many intelligent people they say oh monad is a very interesting way of abstraction in programming just like i have to learn that and at some point i think when i fail understanding and i say oh if i have if i if i try to make a video on it i will finally understand it so usually making videos on something it actually pushes you to understand it i don't know why this is true but i find it very interesting because our brain of course brings a lot of information and uh self-attention it always tells you this is not interesting this is not interesting it has a lot of a big selectivity so this part of selectivity tells you which direction you go so when you make a video on something it will actually now say okay now i i know this is important so it gives you it it makes your attention on on the topic without that it will always something tells you to procrastinate and not not pay attention to it so the reason why i make videos is just interest to be quite honest i make videos when there is some idea and i like to represent i like to learn about it and i like to represent that in a video uh so one of the these ideas which i like local maxima idea i i find it fascinating and actually it links back to computational irreducible in some way because uh local maxima tells me that because we are limited because our brain is limited we we every now and then we will be trapped on a local maxima and that will be very difficult when we drop on a local maxima that's it we we have to become conservatives there is no way around either you risk something some people risk their lives go outside otherwise local maxima the direction will be lost when you when you are trapped so that idea was very interesting and particle simulations i was trying to explain to some friends how because usually when you discuss evolution theory they will tell you how pair cells are arise and i thought i i almost thought the pair cells are actually easy to explain i mean people usually think that the first cells are the complicated cells like now we see no the pair cells are probably just a membrane and the nucleus and i thought that this is very easy to simulate with some particles attracting each other and i was looking at youtube videos to see and then i came across this particle life system uh i don't remember the name of the guy but he was explaining uh these uh cellular shapes with attraction and repulsion force and i thought wow this is very interesting actually i've tried to implement that by myself and i couldn't improvise that and i was surprised i was trying to do that and then a bug on my screen appeared a flying bug so that was very very surprising i was shocked i mean how

how can i i i'm just randomly examining some simple rules and i see something like a flag and then i actually uh made made a video and that video actually uh have a lot of views and the code have a lot of participants they made the code way more sophisticated so now it has a lot of gui parts thanks to the participants uh and i think i think it's uh the most uh uh uh uh yeah i mean many many people actually become more popular uh the particle life uh after that video and the reason why i think it's because the code was the simplest possible code i thought i thought because it is educational i have to

make it the simplest possible i i actually reduced all the parts that i thought uh it may not be interesting and because the code is very simple and the patterns is very complex i thought that's what attracted and i am hoping more people actually come to the particle life because i know game of life is very interesting it's turing complete it's very but particle life is less known about but they are more biologically looking yeah a lot a lot a lot of insights there and about the formation of well first the joy and the fun of open source and of people's participation in science around the world and then in the first cells forming there's like hydrophobic and hydrophilic and then exactly that's that was my question about how cute so how so how the material gets there is is you know these are all inquiry to to explore and then also just just about the videos so one thing about videos it's it's an audio and a visual component now some videos you can watch with no audio and and vice versa um but when

audio and visual are happening together it's kind of like two touches from across the markup blanket like if someone's doing hand gestures while they're talking then they're both being emitted from the same cognitive state whereas if it scrambled what hand gestures were happening with the audio it would remove that meaning but synchronously they're experienced together so then they're seen as meaningful so then like i was paying attention to the presentation and it was like slide after slide i would be initially surprised and then through through the suppression of surprise i i keep seeing in in these two images here it's like sometimes you connect with the layer above or below which is top down from the top's perspective or bottom up from the bottom's perspective or you go sideways and so that could be in a grid world lateral spatial that could be through time like on a timeline it's like those are the local moves that can be made you can move up or down a layer of abstraction and generalization and then you

can move around in space and time and cause it's like those are our that's like the push-up and the pull-up of the local interactions these are the moves the dance moves and then the question is just like what what does the party look like and that's the particle simulation what really look what what starts to happen when all of these moves what if some of the particles can do abstraction yes yeah um but can i show you a demo another oh yeah go for it okay so uh so so the this lattice filter i i i do have a video another video on it but i have also a demo so i i i can show you a python code uh so let me open up my uh so uh do you code in python i thought i saw you code in python what what ide do you use um i i use a cursor for the ai augmentation or just vs code oh visual studio code i use uh what what's called spider oh yeah for for for scientific python yes it it it is very similar to matlab because in my phd i i left matlab i i i hate it as a language but it's very nice ive it has a very nice ive hate it as a language love it as a friend exactly so when i trans transitioned to python code i saw spider as a home because it looked

like uh it looked like uh it looked like uh the matlab the yeah also like r studio oh yeah yeah

okay so so you see them ide now huh yep go for it great

so let me

so let me

so

okay so i i'm pasting a previous

okay

let me paste a quote from

okay

so can you see uh

can you see the code okay yep yep okay so so this is i i like this algorithm uh i people usually call it lattice filters uh as i said i derived this algorithm just from the prediction coding uh

multilinear regression system and i have a video about that uh it's titled regression

and brain so it's very very simple code but it somewhat explains repetition suppression

so

if you if you run the code okay so now this this will act this is activity this is the error and this is this by the way do you see the plots or you just see the spider ive no plots just the ive okay so i have to share the screen the screen is better so let me share the screen

i think you should see the plot oh yeah all right looks good okay great so i run the code uh and this is the pattern so so let's let's simulate a sinusoidal pattern in the beginning so so let's say let's say we are just simulating a sinusoidal

okay

so this is the truth the green and the and the orange is the prediction you see in the beginning it doesn't predict the the the orange line doesn't overlaid the blue line the the sinusoidal activity so the sinusoidal activity is our data the truth and and the prediction is that is the algorithm

which i showed you it's just it's just few lines of code here so i will just explain it right now but

but but the interesting connection i will try to make to uh repetition suppression so it's see you see

after few iterations it can perfectly predict the the sinusoidal activity uh and and you see the error plot actually reducing in the beginning it's high but then it's reducing uh to zero and and this actually tells you how much activity do we have in different layers

so you can see in the beginning in the lower layers we have higher activity let me zoom in so we have higher activity in the lower layers and then as we go to the upper layers we have less activity that's why because because the system can predict what happens now i will introduce a surprise to this system

let me uh in this code now i am for every for every two steps i am just changing y to one it means just sinusoidal wave but i'm introducing one to it so it's just like now we have a surprise okay now you see

it's it's more difficult so in the beginning it can't predict the entire system but then later on it

it learns it learns to predict the sequence so you see it's just the same sinusoidal wave but i have

introduced a number one uh i have introduced something surprising uh but then later on it predicts

and then the error goes down and then again we have the activities uh but now the activities can go to higher layers before it's suppressed to zero okay now i will just introduce one surprising event

so just minus one at at point 17. so at point 17 i will just introduce a surprise to the sequence

and see what the algorithm how the algorithm behaves okay so you see

okay so you see it predicts the system but here when i introduced a surprise it can't predict it it can't fit the data and and if you see the activity you see when the at the level at this level when i introduced the surprise you can see the prediction error goes all the way all the upper layers so just like repetition suppression when you see something surprising the activity increases so just like that in the beginning so this algorithm is actually a very nice view to tell us how our brain may be working and here we we even see that that almost diagonal propagation exactly up into the right as it sort of normalizes and calms down at higher levels exactly exactly so so maybe maybe that's a good thing and a bad thing because they're going to learn this very little surprise that's why even in the upcoming uh steps it needs some time to come back so maybe maybe maybe we can put a filter there we can say oh if the prediction error is very high don't learn the pattern so maybe we can put a threshold and maybe that will actually tell us that's why we have belief system in us that's why our brain is actually fixed on some ideas sometimes if something very surprising and it doesn't conform with your previous beliefs because the learning algorithm inside our brain is very fast if we can learn everything that even if doesn't conform to our previous beliefs it means it can mess up sometimes it's similar to catastrophic forgetting it's just like the activity the new activity may superimpose to the previous activities in your brain i can mess up everything and that's why maybe some people are actually hard to change their minds even after if when they say new data when they say new things uh people usually if it doesn't conform with their beliefs they will just deny it because sometimes you need to fix that you need for this surprise to become a partner so that your brain surrenders says okay i want to learn that that's what i think i mean i think this algorithm can tell us more about our brain that we even uh so so it's very very easy it's just it's just like the prediction uh your system so so you have a sequence of data y is and then you just have just focus on these two lines because everything happens here so you have so you just have a forward prediction and backward prediction basically w and this is the way uh multiplied by the activity current activity it will try to predict the future activity $\hat{y}$ and the subtraction uh the difference between these two is the prediction error so if that that prediction error will go to the upper level so we have a nice illustration on that so so this is uh the the code just illustrates this so y_5 here the space tries to predict y_4 and then the prediction error goes up and then y_4 tries to predict Wi-Fi the prediction the the forward prediction and the backward prediction so the prediction error we have to prediction the forward prediction error and the backward prediction error so each level tries to predict the previous prediction error and each prediction error will go up hierarchy and hopefully at the higher hierarchy you will have less prediction error it means like your brain can predict everything i'll just add a note on prediction error the prediction error is denominated in the units of the measurement like if you expect the room to be 30 degrees and it's 31 then the error is one and then the move in active inference is to translate that error signal into information units bits surprise and then surprise if you can compute it exactly for a simple system go for it but then in cases where it's intractable or you want to do an optimization on bounding surprise that's variational free energy and that's been used for decades

variational auto encoder all these other kinds of variational methods and physics so those are all kind of aligned like if you bound if the error were zero the surprise would be minimal the free energy would be minimal if there's a large error the surprise is higher the variational free energy is higher so like they're slightly different formulations but they're all monotonically related they are related i i think they are using other optimization algorithms to learn these so they may be better optimization that's what i say that you have to learn graduate the problem with this system it learns it try to exactly learn and that's why when you show something surprises uh prediction explodes well that that kind of even comes back to the earlier numbers that you showed with the decimal points it's like um are you trying to predict to one significant figure that's coarse graining to just one digit but then if you give a computer program a float precision number it might have a ton of decimal points and then it will think that those are just as important to compute as the earlier decimal points and then it gets catastrophic because it's doing really well on like the the 50th decimal point but it lost the plot on the main thrust oh i think sometimes precision error uh by the way floating you know floating point numbers are also a form of course draining because we can't actually simulate real numbers the real numbers have infinite precision so so the moment we say we use floating point it means we have four screened the real number there is no way without course draining even in in the computers even if you see if you use double precision it's it's just you have at some point you just cut the the number yeah um cool so what are your next research or or directions that you're going to be exploring okay there are two directions one is i actually want to explore more free energy what happens in these particle systems but not just

intuitively because to be quite honest the things that i showed you some some stuff that i thought it is trivial we don't need to even stimulate it uh but i i have to simulate that because sometimes simulations they you surprise just like the first example which i showed you i mean you you think this happens because of this but sometimes this is not true uh the next thing i am actually working on a different algorithm i call it fun predictive hebian unified neurons uh i am so far i am not successful at this algorithm we can see this algorithm is a

variant of this algorithm which i just showed you uh uh but but i am trying to make these algorithms more local i mean learn locally uh the problem is because you so so what's the problem with the current uh neural network approach the neural network works very good we saw magic with gpt but the problem is is these systems they learn very fast very slow uh training they are sample inefficient they you have to give them so many samples so that they can learn the reason why because they you have to tweak the learning

rate to to lower levels because if you tweak the learning rate higher if the system learns fast it will have a catastrophic forgetting the moment it sees something new novel it's because it has high learning rate it tries to incorporate that new thing into all the systems all the neural network nodes and because these systems are distributed it means it tries to fine-tune every node every weight and that could be catastrophic especially if this outlier is just a noise it's just a noise you need to uh ignore but if you have a high learning rate the algorithm doesn't can't learn that's why you have to make the

learning rate small but if you learn make the learning rate small it means you have to iteratively learn so many samples and variations and that's why only big companies nowadays can train neural networks

because it needs so many iterations and even these big companies they can't make a real-time system so even if they make a robot that robot can't learn real time it's just like the company can tell you oh this robot can only learn do things that we have trained in in 2020 you have we have to take it back and update and retrain again on on the data from 2023 or something like that so it's not a real-time learning system so even charging now it's not a real-time learning system but if if opening i need to retrain it again it needed it updated it needed to retrain everything again so basically it means these algorithms are not real time but this algorithm is real time the one i showed you it means once you see the data in just few samples it learns but again the problem with this algorithm is this catastrophic disturbance i mean this instability once it sees something surprising because it tries to learn it if it just mess up and that's uh that's what i try to figure out something that learns fast but it doesn't disturb the entire system so i'm actually trying to do that uh but so far not very successful but but every now then i i have ideas to implement and then try again and again uh i think now i i i mean few months ago i have a new eye but i still don't have time to implement it so maybe maybe you can hear more about me in the future about that predicted hidden unified neurons which i shortly call it fun which which is not fun

it doesn't work it's not fun yet

planning for fun is still fun um okay do you have any last comments uh no thank you very much for actually inviting me inviting me to to this uh and i hope you continue i actually find active inference channel very interesting and i find the topic is very interesting actually the entire topic that you specialized yourself to just this one topic and it's a very very interesting i i thought is a very good candidate for the theory of everything i mean people usually think that the theory of everything should come from physics but i think it will come from neuroscience that's my thing i mean it doesn't make sense to have a theory of everything without consciousness that's why it has to come from neuroscience i mean physics can help in neuroscience but the theory of everything it comes from neuroscience there is no way that's my or it comes from a new combinatoric fusion i mean one of the key moves in the free energy principle is to talk about the cognitive particle so then if it's a bayesian mechanics if it's a statistical physics for cognitive particles then it's not surprising that it has a lot to do with particle physics

yes yeah yeah maybe maybe i come here that that's the best thing to

to actually do to actually do i actually did i mean i have maybe uh one extra uh animations like i i i can show you before sounds good yes about this combinatoric things i usually show that to to friends and and they usually like it and i explained that in in in my videos in some of my videos

so i mean if you if we if we see this graph as the entire human knowledge and then this child it likes to learn it's just born and it doesn't know anything so the problem is our age is very limited so by the time that the the child the curious child that tries to collect information it will get old and it will

at some point it will die before learning about everything and that that's the reality okay now what i think actually a better is specialization it means that child shouldn't try to learn everything in the beginning it tries to learn all the things that are surrounding and then should specialize if you specialize at some point you will reach the edge of knowledge and once you reach the edge of knowledge this is where you can find a new information the the extrapolation that's what i call extrapolation okay but the problem with specialization is we will be trapped at a local maxima it means so it means if this guy is a doctor uh it tries to treat cancer it specializes in cancer but maybe the true treatment of cancer comes from an engineering engineering trick someone can find a new radiation that doesn't affect normal cells only a big uh cancer cells so i mean if you trap yourself just reading medicine you can never find this solution uh so so this solution how how it comes so let's say uh so so that's my point you can never make distant connections don't get connections because you specialize you you

trapped yourself now if if other people who are also specialized in different topics uh if they collaborate with each other that's why i'm coming back to you so if physics and collaborate with chemistry with chemists and they collaborate with neuroscientists and then this collaboration we can overcome the local maximum traps totally agree and they're there they they are working on the interdisciplinary team sending each other papers sending each other jokes and memes that reflect their world views to give a kind of qualitative sense of their understanding and also i i think in the course grading i think about nestmate ants returning with food and so they're not saying what temperature or anything it's just interacting with nesme yes there was food someone else came that way with food and so that's like a binary coarse graining that tells you something about that direction and so it's like every node is a local sense maker receiving and sending and that's active inference yes yes that's a nice way to complete okay unar thank you good luck with your with your work and with your education so it's been awesome thank you very much daniel see you soon bye see you bye thank you

Thank you.